Lesson 8.3 Guided Notes

Session 1 · Multiplication Equations

I can solve a multiplication equation by dividing both sides by the same number. · 6.EE.B.7

Word / Spanish	Meaning — complete the blank	Example
coefficient / coeficiente	The number that _____ the variable.	the 3 in 3x
inverse operations / operaciones inversas	× and ÷ _____ each other.	5 × 9 = 45, so 45 ÷ 5 = 9
Division Property of Equality / propiedad de división de la igualdad	Divide both sides by the same number (not _____). They stay equal.	both sides ÷ 3

Key idea

Division _____ multiplication.

To solve 5b = 45, divide both sides by the _____, 5.

Worked example

A room holds 84 chairs in 6 equal stacks. Solve 6c = 84.

The coefficient is _____.

Divide both sides by 6: c = 84 ÷ 6 = _____.

Check: 6 × 14 = _____.

Answer: c = _____ chairs in each stack

You try

1. Solve each equation.
a. 7x = 63 _____
b. 4n = 52 _____

_____

2. Solve 4w = 3.2. Check your answer.

_____

Sum it up: To solve a multiplication equation, I ___ because ___.

_____

Session 2 · Division Equations

I can solve a division equation by multiplying both sides by the same number. · 6.EE.B.7

Word / Spanish	Meaning — complete the blank	Example
divisor / divisor	The number you _____ by.	the 4 in n ÷ 4
Multiplication Property of Equality / propiedad de multiplicación de la igualdad	Multiply _____ sides by the same number. They stay equal.	both sides × 4
equal groups / grupos iguales	Groups that each have the _____ amount.	24 is 4 groups of 6

Key idea

Multiplication _____ division.

To solve n ÷ 4 = 7, multiply both sides by _____.

Worked example

Nine hikers share a bag of trail mix equally. Each gets 7 ounces. Solve t ÷ 9 = 7.

Multiply both sides by _____.

t = 7 × 9 = _____.

Check: 63 ÷ 9 = _____.

Answer: The bag holds _____ ounces.

You try

1. Solve each equation.
a. x ÷ 3 = 12 _____
b. n ÷ 7 = 8 _____

_____

2. Solve n ÷ 2 = 1.5. Check your answer.

_____

Sum it up: To solve a division equation, I ___ because ___.

_____

Session 1 · Model, reason, and explain

Multiplication Equations · 6.EE.B.7

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Six equal groups contain 42 objects. Let g be objects per group. Write, solve, and check a multiplication equation with a tape model.

Story quantities	Equal groups / balance	Equation	Substitute to check

_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: multiplies by the coefficient instead of dividing or expects only whole-number solutions.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Division Equations · 6.EE.B.7

Success: represent the quantities, justify the relationship, and check the result with the correct units.

A total t is shared among 5 groups, each receiving 9 items. Write and solve a division equation. State why the total exceeds one share in this context.

Story quantities	Equal groups / balance	Equation	Substitute to check

_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: divides again or makes an unconditional total/share comparison.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?